King Island Renewable Energy Integration Project

$5.95m
Funded by ARENA

$17.69m
Total project cost

Project overview

Lead Organisation
Hydro Tasmania

Location
King Island, Tasmania

ARENA Program
Advancing Renewables Program

Start date
20 February 2010

End date
19 April 2016

Project Partners
Tasmanian Government

This hybrid technology project was completed on 19 April 2016.

Summary

This project demonstrates a world-leading power system that combines several renewable energy technologies, smart tech integrations and energy management technologies. The system will supply over 65% of King Island's annual energy needs using renewable energy, reducing CO2 emissions by more than 95%.

Need

Communities like King Island, which are not connected to the electricity grid, rely heavily on diesel-generated electricity. It is difficult to integrate large amounts of renewable energy into such power systems without impacting on reliability or supply. Overcoming these integration problems will significantly reduce this community's dependence on diesel, lower energy production costs and reduce carbon dioxide emissions.

Project innovation

This project demonstrates a world-leading power system that will supply over 65% of King Island's energy needs using renewable energy, thereby reducing carbon dioxide emissions by more than 95%.

The way in which the technologies are used and integrated is world-leading and will provide a reliable and stable electricity supply using a high proportion of renewable energy.

This project consists of:

- Hybrid technologies (Primary)
- Battery storage
- Solar energy
- Solar PV R&D

Project Knowledge

- ARENA media release 03/03/15

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The new power system is a unique combination of several technologies. The renewable energy technologies being used are well-established – wind, solar (PV) and bio diesel. These are combined with new and emerging enabling and storage technologies. The system includes battery energy storage, a diesel-based Uninterruptible Power Supply (D-UPS), a smart grid system and an advanced control system.

The inclusion of the smart grid system helps to match the island's energy needs with the available renewable energy supplies.

Benefit

The project allows the power system to rely less on diesel generation and provide a reliable and stable electricity supply while significantly reducing CO2 emissions.

The project increased awareness in other communities with similar off-grid systems on how renewable energy sources and enabling technology can provide reliable electricity generation. the lessons from this project have been directly incorporated into the Flinders Island Hybrid Energy Hub and Rottnest Island Water and Renewable Energy Nexus Projects.

Read more

- King Island project website

Last updated 10 February 2021

ARENAWIRE Blogs

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